## Symptom:

**P0116-ENGINE COOLANT TEMPERATURE CIRCUIT PERFORMANCE**

### When Monitored and Set Condition:

**P0116-ENGINE COOLANT TEMPERATURE CIRCUIT PERFORMANCE**

When Monitored: Engine off time is greater than 480 minutes. Ambient temperature is greater than 4 deg C (38 deg F).

Set Condition: After a calibrated amount of cool down time the PCM compares the ECT Sensor, IAT Sensor, and the Ambient Air Temp Sensor values. If the IAT Sensor value is not within 10 deg C (18 deg F) of the other two temperature sensors. Two Trip Fault.

| POSSIBLE CAUSES |
|---|
| GOOD TRIP EQUAL TO ZERO |
| (K2) ECT SIGNAL CIRCUIT SHORTED TO BATTERY VOLTAGE |
| ECT SENSOR |
| (K2) ECT SIGNAL CIRCUIT OPEN |
| (K900) SENSOR GROUND CIRCUIT OPEN |
| (K2) ECT SIGNAL CIRCUIT SHORTED TO GROUND |
| (K2) ECT SIGNAL CIRCUIT SHORTED TO THE (K900) SENSOR GROUND |
| PCM |

| TEST | ACTION | APPLICABILITY |
|---|---|---|
| 1 | Ignition on, engine not running.<br>With the DRBIII®, read DTCs and record the related Freeze Frame data.<br>Is the Good Trip Counter displayed and equal to zero?<br>Yes → Go To 2<br>No → Refer to the INTERMITTENT CONDITION Symptom (Diagnostic Procedure).<br>Perform POWERTRAIN VERIFICATION TEST VER - 5. | All |
| 2 | Turn the ignition off.<br>Disconnect the C2 PCM harness connector.<br>Disconnect the ECT Sensor harness connector.<br>**NOTE: Visually inspect both the component and the PCM connectors. Look for damaged, partially broken wires, and backed out or corroded terminals.**<br>Ignition on, engine not running.<br>Measure the voltage on the (K2) ECT Signal circuit at the ECT Sensor harness connector.<br>Is the voltage above 5.2 volts?<br>Yes → Repair the short to voltage in the (K2) ECT Signal circuit.<br>Perform POWERTRAIN VERIFICATION TEST VER - 5.<br>No → Go To 3 | All |